

Lab 2

DE Genes Enrichment Analysis

Jiaxing Chen

Tasks

1. In R workspace, load the dataset by:

```
rawCounts <- read.delim("Lab2/E-GEOD-50760-raw-counts.tsv")
```

```
# Read in the sample mappings
```

```
sampleData <- read.delim("Lab2/E-GEOD-50760-experiment-design.tsv")
```

2. Read the manual in <https://bioconductor.org/packages/release/bioc/html/DESeq2.html>, then run DESeq2 to identify differential expressed (DE) genes between tumor and normal. List the DE genes by p-value (save as .csv file).

3. Conduct gene set enrichment to the identified DE genes. List the function that domain the difference between tumor and normal. Describe your conclusion with the table and figures generated by gene set enrichment.

4. Apply hierarchical clustering to genes, then draw the cluster dendrogram and heatmap. (Ref https://tavareshugo.github.io/data-carpentryrnaseq/04b_rnaseq_clustering.html)

5. (Optional) Conduct gene set enrichment to each gene cluster.

Data Structure

- E-GEOD-50760-raw-counts.tsv

Gene.ID	Gene.Name	SRR975551	SRR975552	SRR975553	SRR975554	SRR975555	SRR975556	SRR975557	SRR975558	SRR975559	SRR975560	SRR975561	SRR975562	SRR975563	SRR975564	SRR975565	SRR975566	SRR975567	SRR975568
ENSG00000000003	TSPAN6	6617	1352	1492	3390	1464	1251	207	1333	2126	1799	1362	3435	1711	487	4326	3652	1090	2460
ENSG00000000005	TNMD	69	1	20	23	12	4	20	2	3	6	10	15	2	3	21	7	3	4
ENSG000000000419	DPM1	2796	714	510	1140	1667	322	273	621	1031	677	480	1194	634	679	1347	849	692	1112
ENSG000000000457	SCYL3	486	629	398	239	383	290	164	452	172	229	264	297	548	232	454	172	412	331
ENSG000000000460	C1orf112	466	342	73	227	193	35	38	184	174	68	46	173	81	194	241	158	223	170
ENSG000000000938	FGR	75	95	158	107	135	75	236	254	121	107	94	90	113	360	84	227	168	95
ENSG000000000971	CFH	803	662	1943	459	383	965	4157	1331	758	1081	864	738	1406	507	393	786	1082	446
ENSG000000001036	FUCA2	2740	1896	1344	2069	2095	986	597	2407	1495	1506	685	2101	1986	1621	1984	1038	1105	1917
ENSG000000001084	GCLC	1823	3015	714	1440	1212	441	469	881	860	593	540	742	869	749	1350	690	1103	995
ENSG000000001167	NFYA	1452	1736	526	1147	969	374	252	784	797	497	289	742	751	723	622	554	1743	609
ENSG000000001460	STPG1	188	196	249	102	294	132	111	332	203	253	211	202	243	125	102	186	182	59
ENSG000000001461	NIPAL3	1244	1211	739	503	1800	682	657	1736	857	762	832	614	1300	983	484	440	790	596
ENSG000000001497	LA51L	2149	1205	553	1176	994	415	479	895	1311	767	394	1364	644	904	1242	1205	664	1059
ENSG000000001561	ENPP4	606	777	622	392	777	370	231	627	439	468	498	547	1114	409	689	290	702	631
ENSG000000001617	SEMA3F	203	941	265	282	1019	281	396	527	1176	722	482	832	691	308	575	851	646	425
ENSG000000001626	CFTR	3477	2306	3421	1233	6421	3305	18	4068	3639	1559	2175	1224	6244	1471	2522	3575	2645	8686
ENSG000000001629	ANKIB1	2032	1709	889	803	1422	682	609	1384	713	582	550	665	1301	671	1212	827	880	1046
ENSG000000001630	CYP51A1	276	190	115	241	314	132	46	265	130	127	125	264	286	341	312	267	117	232
ENSG000000001631	KRIT1	1471	972	466	724	745	416	390	709	439	306	337	472	775	456	1230	467	479	599
ENSG000000002016	RAD52	253	298	289	172	209	146	86	194	109	174	111	150	264	170	164	124	188	159
ENSG000000002079	MYH16	7	4	3	3	5	5	1	20	2	7	2	8	3	4	21	3	2	6
ENSG000000002330	BAD	510	419	410	221	572	475	250	660	449	851	521	408	1181	424	165	197	374	244
ENSG000000002549	LAP3	2538	1762	2294	1459	3723	1126	1696	3881	1825	1482	1198	1295	2087	2872	2564	2473	2108	1663
ENSG000000002586	CD99	6729	1193	1623	2845	1777	2033	3035	2801	4513	2761	2084	3349	2821	2395	2997	2038	2407	1392
ENSG000000002587	H535T1	118	136	133	273	186	94	67	425	158	157	107	81	216	251	284	115	170	104
ENSG000000002726	AOC1	3365	2555	6283	1545	3077	6084	24	4621	1340	7181	4531	3157	9593	1314	2854	1708	2890	2010
ENSG000000002745	WNT16	0	0	1	4	0	1	12	10	2	1	2	1	2	31	2	1	2	1

Data Structure

- **E-GEOD-50760-experiment-design.tsv**

Run	Sample.Characteristic.biopsy.site.	Sample.Characteristic.Ontology.Term.biopsy.site.	Sample.Characteristic.disease.	Sample.Characteristic.Ontology.Term.disease.	Sample.Characteristic.disease.staging.	Sample.Characteristic.Ontology.Term.disease.staging.	Sample.Characteristic.individual.	Sample.Characteristic.Ontology.Term.individual.
SRR975551	primary tumor	http://www.ebi.ac.uk/efo/EFO_0000616	colorectal cancer	http://www.ebi.ac.uk/efo/EFO_0005842	Stage IV Colorectal Cancer	NA	AMC_2	NA
SRR975552	primary tumor	http://www.ebi.ac.uk/efo/EFO_0000616	colorectal cancer	http://www.ebi.ac.uk/efo/EFO_0005842	Stage IV Colorectal Cancer	NA	AMC_3	NA
SRR975553	primary tumor	http://www.ebi.ac.uk/efo/EFO_0000616	colorectal cancer	http://www.ebi.ac.uk/efo/EFO_0005842	Stage IV Colorectal Cancer	NA	AMC_5	NA
SRR975554	primary tumor	http://www.ebi.ac.uk/efo/EFO_0000616	colorectal cancer	http://www.ebi.ac.uk/efo/EFO_0005842	Stage IV Colorectal Cancer	NA	AMC_6	NA
SRR975555	primary tumor	http://www.ebi.ac.uk/efo/EFO_0000616	colorectal cancer	http://www.ebi.ac.uk/efo/EFO_0005842	Stage IV Colorectal Cancer	NA	AMC_7	NA
SRR975556	primary tumor	http://www.ebi.ac.uk/efo/EFO_0000616	colorectal cancer	http://www.ebi.ac.uk/efo/EFO_0005842	Stage IV Colorectal Cancer	NA	AMC_8	NA
SRR975557	primary tumor	http://www.ebi.ac.uk/efo/EFO_0000616	colorectal cancer	http://www.ebi.ac.uk/efo/EFO_0005842	Stage IV Colorectal Cancer	NA	AMC_9	NA
SRR975558	primary tumor	http://www.ebi.ac.uk/efo/EFO_0000616	colorectal cancer	http://www.ebi.ac.uk/efo/EFO_0005842	Stage IV Colorectal Cancer	NA	AMC_10	NA
SRR975559	primary tumor	http://www.ebi.ac.uk/efo/EFO_0000616	colorectal cancer	http://www.ebi.ac.uk/efo/EFO_0005842	Stage IV Colorectal Cancer	NA	AMC_12	NA
SRR975560	primary tumor	http://www.ebi.ac.uk/efo/EFO_0000616	colorectal cancer	http://www.ebi.ac.uk/efo/EFO_0005842	Stage IV Colorectal Cancer	NA	AMC_13	NA
SRR975561	primary tumor	http://www.ebi.ac.uk/efo/EFO_0000616	colorectal cancer	http://www.ebi.ac.uk/efo/EFO_0005842	Stage IV Colorectal Cancer	NA	AMC_17	NA
SRR975562	primary tumor	http://www.ebi.ac.uk/efo/EFO_0000616	colorectal cancer	http://www.ebi.ac.uk/efo/EFO_0005842	Stage IV Colorectal Cancer	NA	AMC_18	NA
SRR975563	primary tumor	http://www.ebi.ac.uk/efo/EFO_0000616	colorectal cancer	http://www.ebi.ac.uk/efo/EFO_0005842	Stage IV Colorectal Cancer	NA	AMC_19	NA
SRR975564	primary tumor	http://www.ebi.ac.uk/efo/EFO_0000616	colorectal cancer	http://www.ebi.ac.uk/efo/EFO_0005842	Stage IV Colorectal Cancer	NA	AMC_20	NA
SRR975565	primary tumor	http://www.ebi.ac.uk/efo/EFO_0000616	colorectal cancer	http://www.ebi.ac.uk/efo/EFO_0005842	Stage IV Colorectal Cancer	NA	AMC_21	NA
SRR975566	primary tumor	http://www.ebi.ac.uk/efo/EFO_0000616	colorectal cancer	http://www.ebi.ac.uk/efo/EFO_0005842	Stage IV Colorectal Cancer	NA	AMC_22	NA
SRR975567	primary tumor	http://www.ebi.ac.uk/efo/EFO_0000616	colorectal cancer	http://www.ebi.ac.uk/efo/EFO_0005842	Stage IV Colorectal Cancer	NA	AMC_23	NA
SRR975568	primary tumor	http://www.ebi.ac.uk/efo/EFO_0000616	colorectal cancer	http://www.ebi.ac.uk/efo/EFO_0005842	Stage IV Colorectal Cancer	NA	AMC_24	NA
SRR975569	normal	http://purl.obolibrary.org/obo/PATO_0000461	colorectal cancer	http://www.ebi.ac.uk/efo/EFO_0005842	Stage IV Colorectal Cancer	NA	AMC_2	NA
SRR975570	normal	http://purl.obolibrary.org/obo/PATO_0000461	colorectal cancer	http://www.ebi.ac.uk/efo/EFO_0005842	Stage IV Colorectal Cancer	NA	AMC_3	NA
SRR975571	normal	http://purl.obolibrary.org/obo/PATO_0000461	colorectal cancer	http://www.ebi.ac.uk/efo/EFO_0005842	Stage IV Colorectal Cancer	NA	AMC_5	NA
SRR975572	normal	http://purl.obolibrary.org/obo/PATO_0000461	colorectal cancer	http://www.ebi.ac.uk/efo/EFO_0005842	Stage IV Colorectal Cancer	NA	AMC_6	NA
SRR975573	normal	http://purl.obolibrary.org/obo/PATO_0000461	colorectal cancer	http://www.ebi.ac.uk/efo/EFO_0005842	Stage IV Colorectal Cancer	NA	AMC_7	NA
SRR975574	normal	http://purl.obolibrary.org/obo/PATO_0000461	colorectal cancer	http://www.ebi.ac.uk/efo/EFO_0005842	Stage IV Colorectal Cancer	NA	AMC_8	NA
SRR975575	normal	http://purl.obolibrary.org/obo/PATO_0000461	colorectal cancer	http://www.ebi.ac.uk/efo/EFO_0005842	Stage IV Colorectal Cancer	NA	AMC_9	NA
SRR975576	normal	http://purl.obolibrary.org/obo/PATO_0000461	colorectal cancer	http://www.ebi.ac.uk/efo/EFO_0005842	Stage IV Colorectal Cancer	NA	AMC_10	NA
SRR975577	normal	http://purl.obolibrary.org/obo/PATO_0000461	colorectal cancer	http://www.ebi.ac.uk/efo/EFO_0005842	Stage IV Colorectal Cancer	NA	AMC_12	NA

Introduction to DESeq2 # Installation & Usage

```
if (!require("BiocManager", quietly = TRUE))
  install.packages("BiocManager") # Installation of BiocManager

BiocManager::install("DESeq2") # Installation of DESeq2
library("DESeq2")
```

```
### Construct DESeqDataSet
dds <- DESeqDataSetFromMatrix(countData = countData,
                              colData = colData,
                              design = ~design)
```

- DESeq2 will not make guesses as to which column of the count matrix belongs to which row of the column data, these must be provided to DESeq2 **already in consistent order**.
- **Order: countData col = colData row**

Introduction to DESeq2 # Data Prepare

	SRR975551	SRR975552	SRR975553	SRR975554	SRR975555	SRR975556	SRR975557	SRR975558	SRR975559
ENSG000000000003	6617	1352	1492	3390	1464	1251	207	1333	2126
ENSG000000000005	69	1	20	23	12	4	20	2	3
ENSG000000000419	2798	714	510	1140	1667	322	273	621	1031
ENSG000000000457	486	629	398	239	383	290	164	452	172
ENSG000000000460	466	342	73	227	193	35	38	184	174
ENSG000000000938	75	95	158	107	135	75	236	254	121
ENSG000000000971	803	662	1943	459	383	965	4157	1331	758
ENSG00000001036	2740	1896	1344	2069	2095	986	597	2407	1495
ENSG00000001084	1823	3015	714	1440	1212	441	469	881	860
ENSG00000001167	1452	1736	526	1147	969	374	252	784	797
ENSG00000001460	188	196	249	102	294	132	111	332	203
ENSG00000001461	1244	1211	739	503	1800	682	657	1736	857
ENSG00000001497	2149	1205	553	1176	994	415	479	895	1311
ENSG00000001561	606	777	622	392	777	370	231	627	439
ENSG00000001617	203	941	265	282	1019	281	396	527	1176
ENSG00000001626	3477	2306	3421	1233	6421	3305	18	4068	3639
ENSG00000001629	2032	1709	889	803	1422	682	609	1384	713
ENSG00000001630	276	190	115	241	314	132	46	265	130
ENSG00000001631	1471	972	466	724	745	416	390	709	439
ENSG00000002016	253	298	289	172	209	146	86	194	109
ENSG00000002079	7	4	3	3	5	5	1	20	2
ENSG00000002330	510	419	410	221	572	475	250	660	449
ENSG00000002549	2538	1762	2294	1459	3723	1126	1698	3881	1825
ENSG00000002586	6729	1193	1623	2845	1777	2033	3035	2801	4513
ENSG00000002587	118	136	133	273	186	94	67	425	158
ENSG00000002726	3365	2555	6283	1545	3077	6084	24	4621	1340

countData

```
> class(countData)
[1] "data.frame"
```

```
##          untreated1 untreated2 untreated3 untreated4 treated1 treated2
## FBgn00000003         0         0         0         0         0         0
## FBgn00000008        92        161         76         70        140        88
##          treated3
## FBgn00000003         1
## FBgn00000008        70
```

countData example

Introduction to DESeq2 # Data Prepare

	condition
SRR975551	primary tumor
SRR975552	primary tumor
SRR975553	primary tumor
SRR975554	primary tumor
SRR975555	primary tumor
SRR975556	primary tumor
SRR975557	primary tumor
SRR975558	primary tumor
SRR975559	primary tumor
SRR975560	primary tumor
SRR975561	primary tumor
SRR975562	primary tumor
SRR975563	primary tumor
SRR975564	primary tumor
SRR975565	primary tumor
SRR975566	primary tumor
SRR975567	primary tumor
SRR975568	primary tumor
SRR975569	normal
SRR975570	normal
SRR975571	normal

colData

```
> class(colData)  
[1] "data.frame"
```

##	condition	type
## treated1fb	treated	single-read
## treated2fb	treated	paired-end
## treated3fb	treated	paired-end
## untreated1fb	untreated	single-read
## untreated2fb	untreated	single-read
## untreated3fb	untreated	paired-end
## untreated4fb	untreated	paired-end

colData example

Introduction to DESeq2 # Data Prepare

```
> design
[1] "primary tumor"
[2] "primary tumor"
[3] "primary tumor"
[4] "primary tumor"
[5] "primary tumor"
[6] "primary tumor"
[7] "primary tumor"
[8] "primary tumor"
[9] "primary tumor"
[10] "primary tumor"
[11] "primary tumor"
[12] "primary tumor"
[13] "primary tumor"
[14] "primary tumor"
[15] "primary tumor"
[16] "primary tumor"
[17] "primary tumor"
[18] "primary tumor"
[19] "normal"
```

design

```
> class(design)
[1] "character"
```

Name	Type	Value
design	character [54]	'primary tumor' 'primary tumor' 'primary tumor' 'primary tumor' 'primary tumor' ...

Running DESeq2 # Build DESeqDataSet

```
### Construct DESeqDataSet, already done in slides 5
dds <- DESeqDataSetFromMatrix(countData = countData,
                              colData = colData,
                              design = ~design)
```

```
> dds
class: DESeqDataSet
dim: 25212 54
metadata(1): version
assays(6): counts mu ... replaceCounts replaceCooks
rownames(25212): ENSG000000000003 ENSG000000000005 ... ENSG00000281904
               ENSG00000281912
rowData names(27): baseMean baseVar ... maxCooks replace
colnames(54): SRR975551 SRR975552 ... SRR975603 SRR975604
colData names(3): condition sizeFactor replaceable
```

- Here, we recommend analyzing **only primary tumor vs normal**.
- If all three datasets are analyzed, it will **also be scored**.

Running DESeq2 # DESeq2 Calculation

```
### Construct DESeqDataSet, already done in slides 5
dds <- DESeq(dds)
res <- results(dds)
```

```
> res
log2 fold change (MLE): condition primary.tumor vs colorectal.cancer.metastatic.in.the.liver
Wald test p-value: condition primary.tumor vs colorectal.cancer.metastatic.in.the.liver
DataFrame with 25212 rows and 6 columns
```

	baseMean	log2FoldChange	lfcSE	stat	pvalue	padj
	<numeric>	<numeric>	<numeric>	<numeric>	<numeric>	<numeric>
ENSG000000000003	1824.881	-0.0729469	0.258834	-0.281830	0.778074	0.934661
ENSG000000000005	10.852	0.2640252	0.439182	0.601175	0.547724	0.835480
ENSG000000000419	725.173	-0.1847973	0.168139	-1.099075	0.271735	0.636548
ENSG000000000457	311.358	0.0987185	0.125400	0.787231	0.431147	0.769127
ENSG000000000460	126.362	-0.1894895	0.233107	-0.812888	0.416282	0.758990
...	...	...	...	...	...	...
ENSG00000281883	1.57310	-0.0221791	0.501402	-0.0442341	0.964718	0.991903
ENSG00000281896	6.70208	-0.0413966	0.278819	-0.1484711	0.881971	0.966623
ENSG00000281903	5.40739	0.1623701	0.297321	0.5461108	0.584990	0.852924
ENSG00000281904	1.32900	0.4728780	0.480022	0.9851167	0.324567	0.686622
ENSG00000281912	10.29742	-0.1508453	0.231623	-0.6512536	0.514883	0.818253

Running DESeq2 # DESeq2 Calculation

```
> summary(res)
```

```
out of 25212 with nonzero total read count
```

```
adjusted p-value < 0.1
```

```
LFC > 0 (up)           : 1491, 5.9%
```

```
LFC < 0 (down)         : 1664, 6.6%
```

```
outliers [1]           : 0, 0%
```

```
low counts [2]          : 0, 0%
```

```
(mean count < 0)
```

Gene Set Enrichment Analysis

- **Gene set enrichment analysis (GSEA)** (also called **functional enrichment analysis** or **pathway enrichment analysis**) is a method to identify classes of genes or proteins that are **over-represented in a large set of genes or proteins**, and may have an association with **different phenotypes** (e.g. different organism growth patterns or diseases).
- **Applications**
 - **GWAS**
 - **Cancer cell profiling**
 - **Depression**
 - **Etc.**

Gene Set Enrichment Analysis # clusterProfiler

- A R package for comparing biological themes among gene clusters, developed by **Prof. Guangchuang Yu, Southern Medical University**.

➤ OMICS. 3.3 【2022最新】 Q2 生物学-3区 2012 May;16(5):284-7. doi: 10.1089/omi.2011.0118.
Epub 2012 Mar 28.

clusterProfiler: an R package for comparing biological themes among gene clusters

Guangchuang Yu ¹, Li-Gen Wang, Yanyan Han, Qing-Yu He

```
### ClusterProfiler Installation
BiocManager::install('clusterProfiler')
library(clusterProfiler)
BiocManager::install("org.Hs.eg.db") # Human genome annotation database
library(org.Hs.eg.db)
```

Gene Set Enrichment Analysis # clusterProfiler

- **clusterProfiler support:**
 - Gene Ontology(GO)
 - Kyoto Encyclopedia of Genes and Genomes(KEGG)
 - Disease Ontology(DO)
 - Disease Gene Network(DisGeNET)
 - Molecular Signatures Database(MSigDb)
 - wikiPathways

clusterProfiler-package	statistical analysis and visualization of functional profiles for genes and gene clusters
append_kegg_category	append_kegg_category
bitr	bitr
bitr_kegg	bitr_kegg
browseKEGG	browseKEGG
clusterProfiler	statistical analysis and visualization of functional profiles for genes and gene clusters
compareCluster	Compare gene clusters functional profile
DataSet	Datasets gcSample contains a sample of gene clusters.
DE_GSE8057	Datasets gcSample contains a sample of gene clusters.
download_KEGG	download_KEGG
dropGO	dropGO
enrichDAVID	enrichDAVID
enricher	enricher
enrichGO	GO Enrichment Analysis of a gene set. Given a vector of genes, this function will return the enrichment GO categories after FDR control.
enrichKEGG	KEGG Enrichment Analysis of a gene set. Given a vector of genes, this function will return the enrichment KEGG categories with FDR control.
enrichMKEGG	KEGG Module Enrichment Analysis of a gene set. Given a vector of genes, this function will return the enrichment KEGG Module categories with FDR control.
enrichPC	enrichPC
enrichWP	enrichWP
gcSample	Datasets gcSample contains a sample of gene clusters.
getPPI	getPPI
getTaxID	getTaxID
getTaxInfo	getTaxInfo
get_wp_organisms	get_wp_organism
Gff2GeneTable	Gff2GeneTable
go2ont	go2ont
go2term	go2term
gofilter	gofilter
groupGO	Functional Profile of a gene set at specific GO level. Given a vector of genes, this function will return the GO profile at a specific level.
groupGOResult-class	Class "groupGOResult" This class represents the result of functional Profiles of a set of gene at specific GO level.
GSEA	GSEA
gseGO	gseGO
gseKEGG	gseKEGG

Gene Set Enrichment Analysis # enrichGO

```
### enrichGO by clusterProfiler  
GO_result <- enrichGO(gene_id_list,  
                        OrgDb = org.Hs.eg.db,  
                        ont = "ALL",  
                        pvalueCutoff = 0.05  
                        pAdjustMethod = "BH",  
                        keyType = 'ENSEMBL')
```

```
> gene_id_list  
[1] "ENSG00000091583" "ENSG00000151655" "ENSG00000171564" "ENSG00000091513" "ENSG00000171557"  
[6] "ENSG00000187758" "ENSG00000117601" "ENSG00000163631" "ENSG00000118520" "ENSG00000145192"  
[11] "ENSG00000198099" "ENSG00000158104" "ENSG00000255974" "ENSG00000118271" "ENSG00000145692"  
[16] "ENSG00000021852" "ENSG00000156096" "ENSG00000198650" "ENSG00000113905" "ENSG00000151224"  
[21] "ENSG00000160868" "ENSG00000175336" "ENSG00000080618" "ENSG00000131482" "ENSG00000072080"  
[26] "ENSG00000138109" "ENSG00000079557" "ENSG00000111713" "ENSG00000132693" "ENSG00000116785"  
[31] "ENSG00000229314" "ENSG00000257017" "ENSG00000122787" "ENSG00000130649" "ENSG00000118137"  
[36] "ENSG00000110169" "ENSG00000104760" "ENSG00000183747" "ENSG00000084674" "ENSG00000055957"  
[41] "ENSG00000171560" "ENSG00000145708" "ENSG00000275152" "ENSG00000105697" "ENSG00000160282"  
[46] "ENSG00000211452" "ENSG00000113600" "ENSG00000121410" "ENSG00000214855" "ENSG00000106927"  
[51] "ENSG00000122194" "ENSG00000145321" "ENSG00000138115" "ENSG00000110245" "ENSG00000162267"  
[56] "ENSG00000188338" "ENSG00000124713" "ENSG00000166035" "ENSG00000109072" "ENSG00000039537"  
[61] "ENSG00000172482" "ENSG00000021826" "ENSG00000113889" "ENSG00000143278" "ENSG00000083807"  
[66] "ENSG00000153086" "ENSG00000099937" "ENSG00000073754" "ENSG00000170509" "ENSG00000145826"  
[71] "ENSG00000100024" "ENSG00000161944" "ENSG00000060566" "ENSG00000277893" "ENSG00000140505"  
[76] "ENSG00000187048" "ENSG00000136881" "ENSG00000175003" "ENSG00000100652" "ENSG00000132703"  
[81] "ENSG00000140107" "ENSG00000251164" "ENSG00000139344" "ENSG00000183549" "ENSG00000241635"  
[86] "ENSG00000009724" "ENSG00000100197" "ENSG00000180432" "ENSG00000171766" "ENSG00000047457"
```

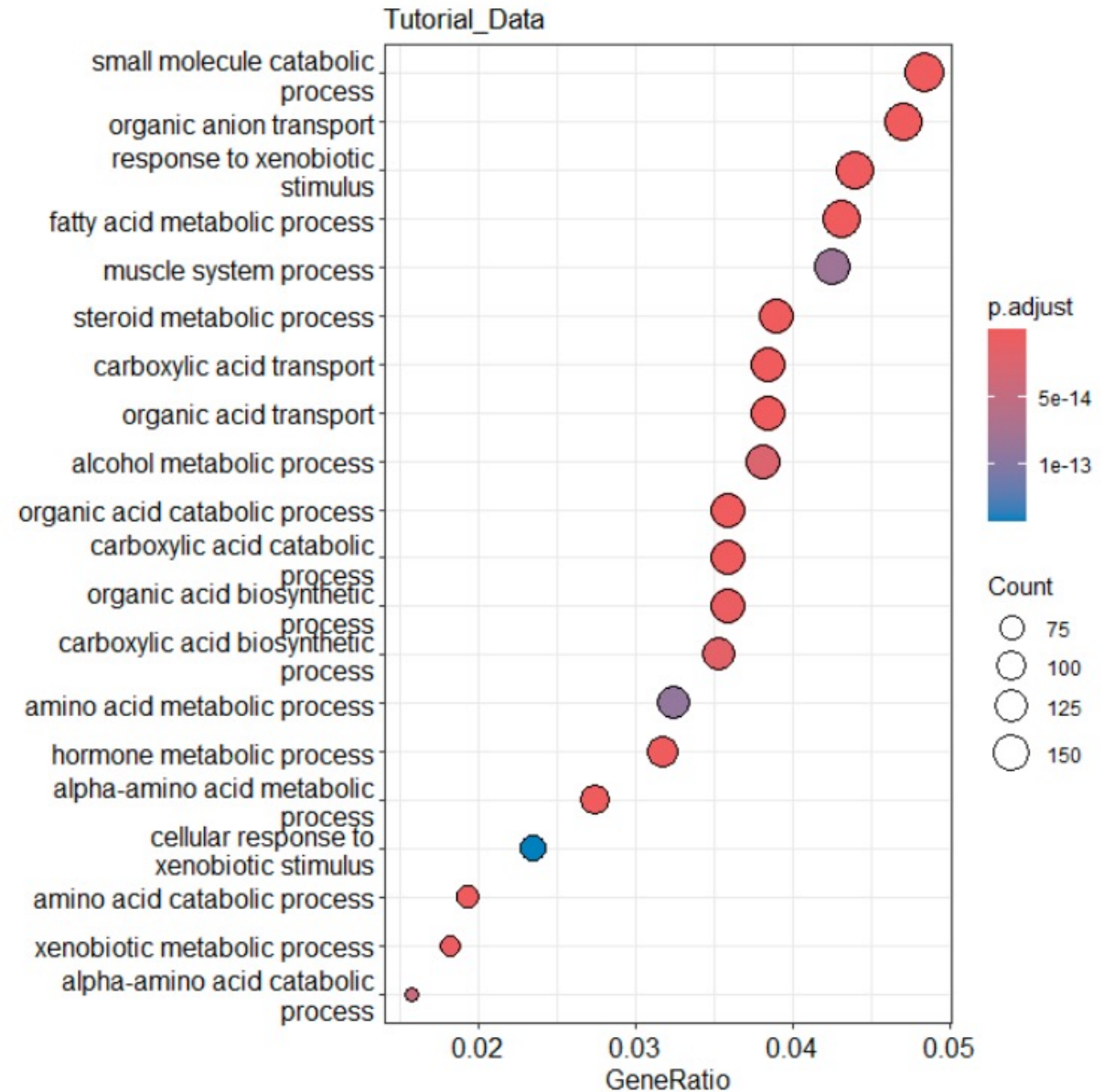
Gene Set Enrichment Analysis # *enrichGO*

```
> class(GO_result)
[1] "enrichResult"
attr(,"package")
[1] "DOSE"
```

```
> GO_result
#
# over-representation test
#
#...@organism      Homo sapiens
#...@ontology      GOALL
#...@keytype       ENSEMBL
#...@gene          chr [1:4988] "ENSG00000091583" "ENSG000000151655" "ENSG000000171564" "ENSG00000091513" "ENSG000000171557" ...
#...pvalues adjusted by 'fdr' with cutoff <0.05
#...1838 enriched terms found
'data.frame':  1838 obs. of  10 variables:
 $ ONTOLOGY      : chr  "BP" "BP" "BP" "BP" ...
 $ ID            : chr  "GO:0044282" "GO:0016054" "GO:0046395" "GO:0008202" ...
 $ Description: chr  "small molecule catabolic process" "organic acid catabolic process" "carboxylic acid catabol
ic process" "steroid metabolic process" ...
 $ GeneRatio     : chr  "173/3570" "128/3570" "128/3570" "139/3570" ...
 $ BgRatio       : chr  "403/21261" "276/21261" "276/21261" "370/21261" ...
 $ pvalue        : num  1.11e-35 1.17e-30 1.17e-30 3.74e-22 8.88e-22 ...
 $ p.adjust      : num  7.11e-32 2.50e-27 2.50e-27 5.99e-19 1.14e-18 ...
 $ qvalue        : num  5.12e-32 1.80e-27 1.80e-27 4.32e-19 8.20e-19 ...
 $ geneID        : chr  "ENSG000000118520/ENSG000000198099/ENSG000000158104/ENSG000000198650/ENSG000000151224/ENSG00000016
0868/ENSG000000122787"| __truncated__ "ENSG000000118520/ENSG000000158104/ENSG000000198650/ENSG000000151224/ENSG00000012
2787/ENSG000000160282/ENSG000000172482"| __truncated__ "ENSG000000118520/ENSG000000158104/ENSG000000198650/ENSG00000015
1224/ENSG000000122787/ENSG000000160282/ENSG000000172482"| __truncated__ "ENSG000000255974/ENSG000000156096/ENSG00000016
0868/ENSG000000175336/ENSG000000131482/ENSG000000138109/ENSG000000122787"| __truncated__ ...
 $ Count         : int  173 128 128 139 168 154 69 113 98 137 ...
#...Citation
T Wu, E Hu, S Xu, M Chen, P Guo, Z Dai, T Feng, L Zhou, W Tang, L Zhan, X Fu, S Liu, X Bo, and G Yu.
clusterProfiler 4.0: A universal enrichment tool for interpreting omics data.
The Innovation. 2021, 2(3):100141
```

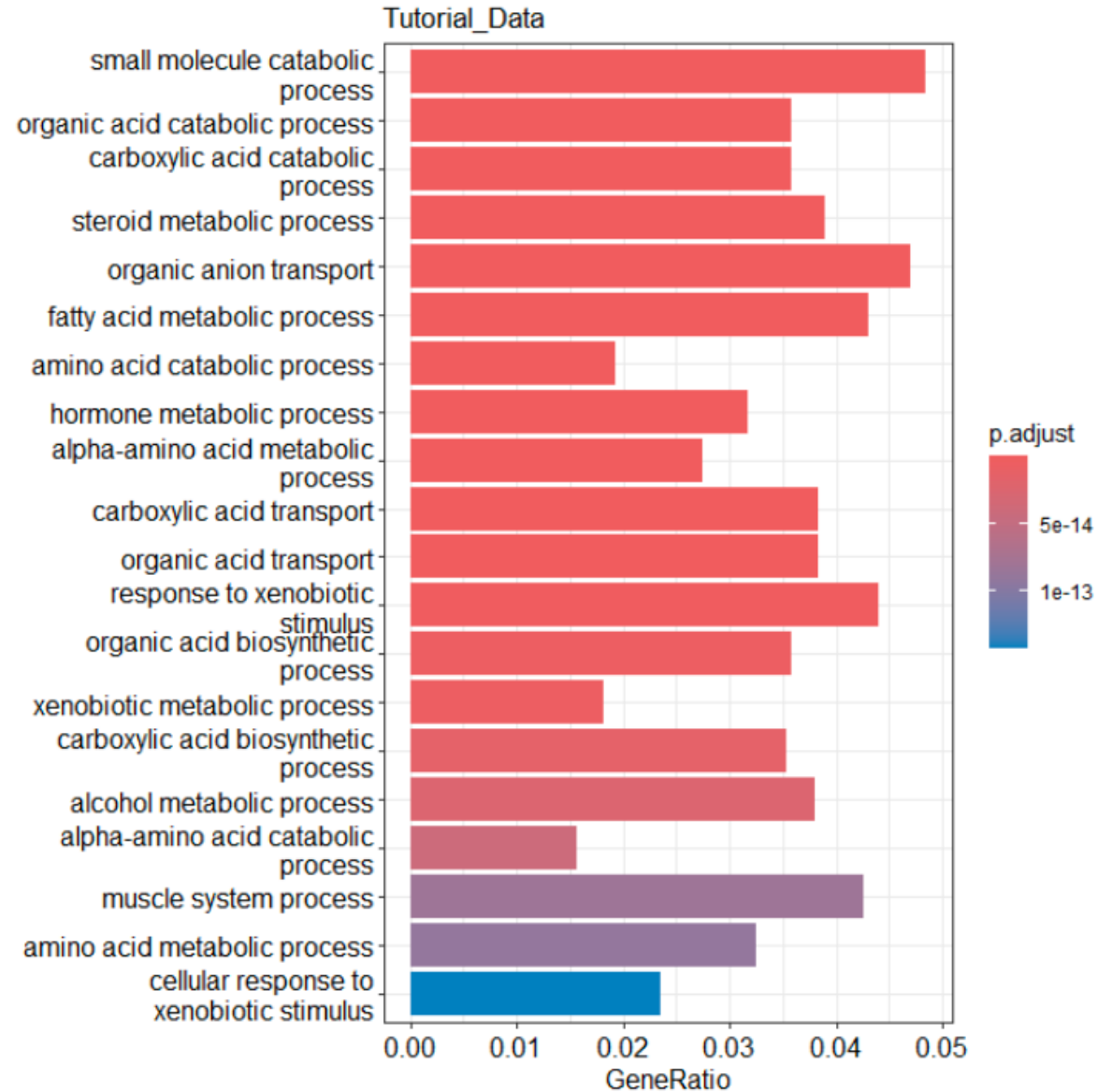
Gene Set Enrichment Analysis # enrichGO Visualization

```
### clusterProfiler visualization  
dotplot(GO_result,  
  x = "GeneRatio",  
  color = "p.adjust",  
  showCategory = 20,  
  size = NULL,  
  title = "Tutorial_Data")
```



Gene Set Enrichment Analysis # enrichGO Visualization

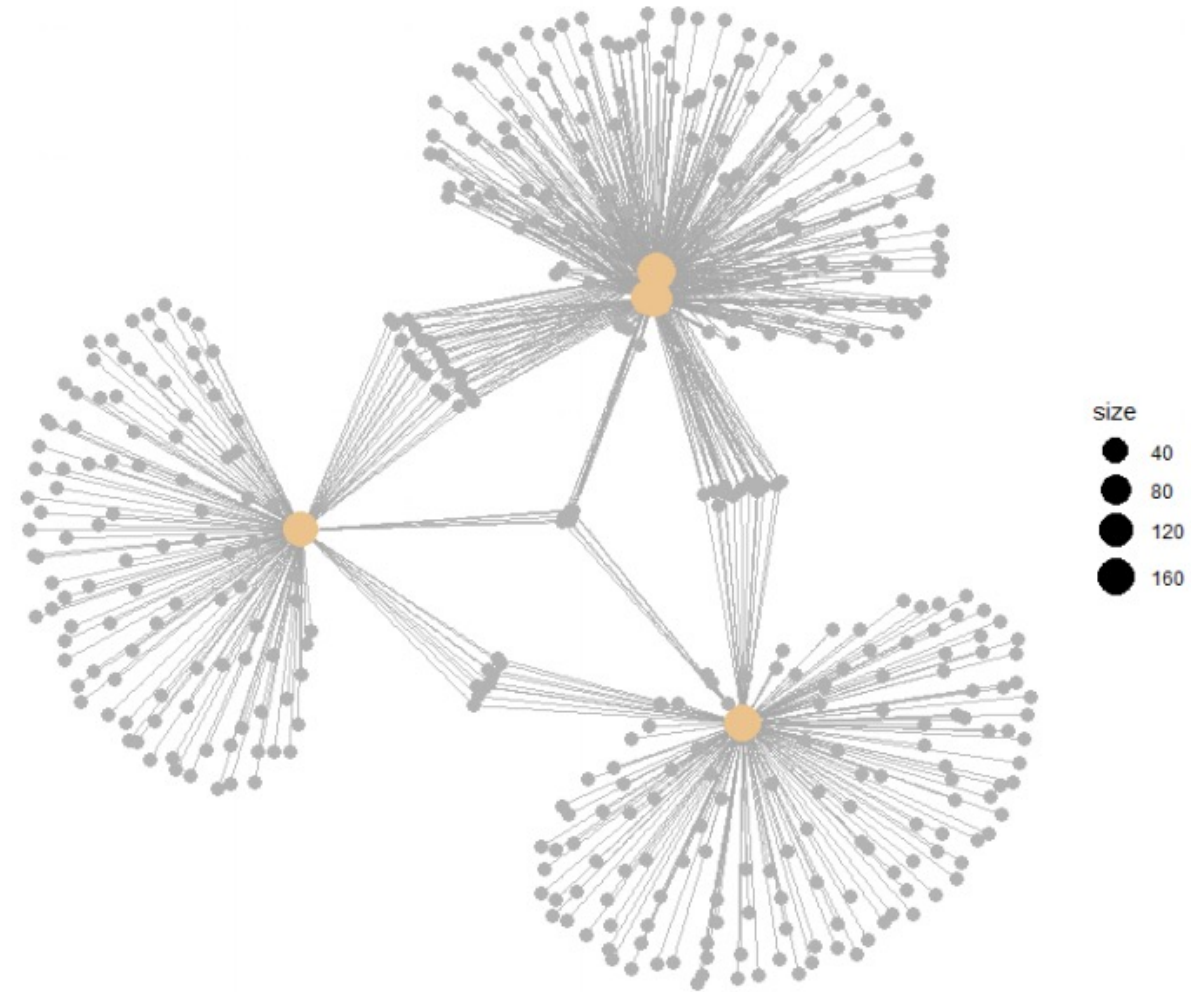
```
### clusterProfiler visualization  
barplot(GO_result,  
        x = "GeneRatio",  
        color = "p.adjust",  
        showCategory = 20,  
        size = NULL,  
        title = "Tutorial_Data")
```



Gene Set Enrichment Analysis # enrichGO Visualization

```
### clusterProfiler visualization  
cnetplot(GO_result,  
         showCategory=5")
```

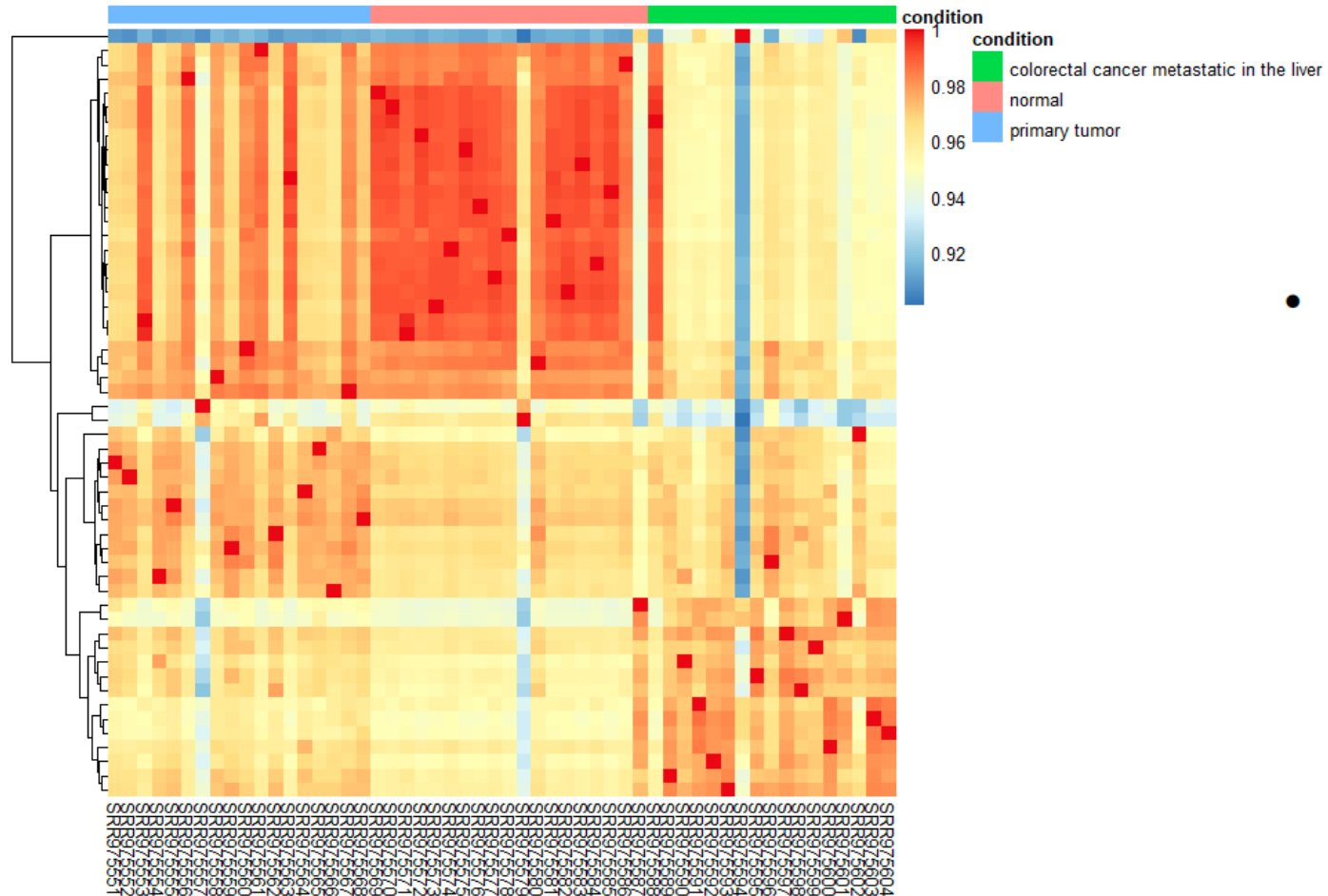
- **Not recommended in this project**
- **enrichKEGG is same as enrichGO**



Gene Set Enrichment Analysis # Heatmap

```
### Install pheatmap  
install.packages("pheatmap")  
library(pheatmap)  
  
pheatmap(cor(count_data), annotation = colData, hcl.colors(50,"BluYl"))
```

Gene Set Enrichment Analysis # Heatmap



- **cluster_cols = FALSE**